

TessTrace

DAG-aware gradient fault localization for composed Tesseract

A wrong vector-Jacobian product can preserve shapes, return finite values, and let optimization converge - while silently producing a worse scientific design.

TessTrace does not merely report a gradient mismatch; it localizes the faulty VJP boundary while preserving healthy sibling branches in a composed Tesseract DAG.

Problem	A final loss curve shows damage, not the faulty boundary.
Diagnosis	UVS is localized as FAIL; Optics is PARTIALLY_CONTAMINATED.
Precision	VS, Human, and Thermal remain PASS.
Effect	Fixing one VJP improves UVS 5.1%, VS 7.5%, and common forward loss.
Evidence	Tesseract 1.11.0, six images, real apply/VJP calls, raw artifact.

Reproducible engineering claim

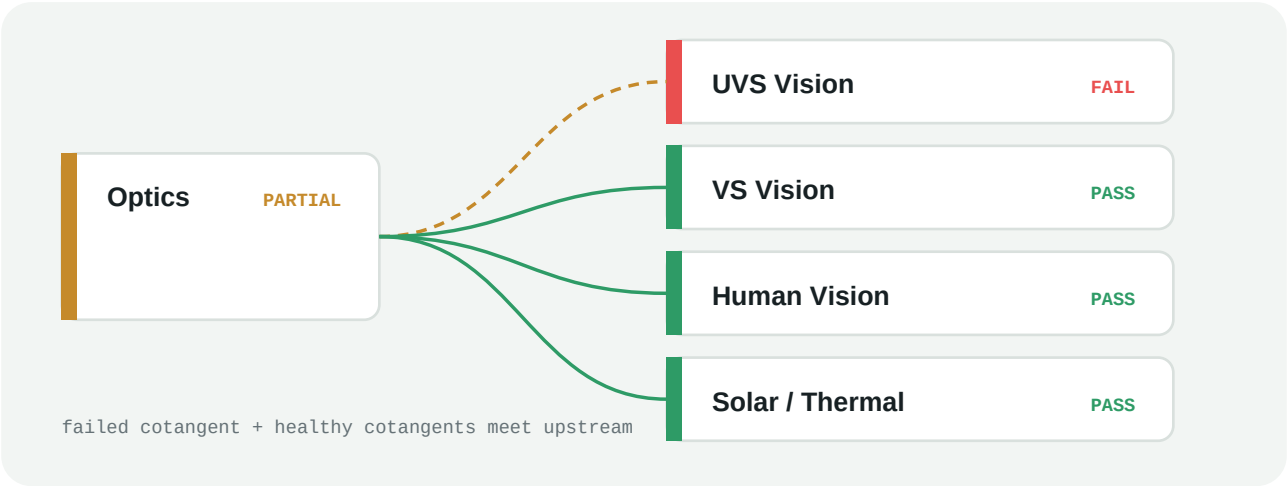
Faulty execution returns code 1; corrected execution returns code 0. CI records the analytic and finite-difference scalars, directional errors, node/edge states, endpoint lists, image SHAs, and workflow URL.

Localize first. Propagate precisely.

For $y=f(x)$, input direction d , and output cotangent c , TessTrace compares:

```
analytic = <VJP_f(x,c), d>
reference = <c, [f(x + epsilon d) - f(x - epsilon d)] / (2 epsilon)>
```

The experiment freezes $\epsilon=1e-6$, $rtol=1e-4$, and $atol=1e-7$ before inspecting results. Each check stores both scalars plus absolute and relative error.



PASS	local contract passed
FAIL	local VJP defect directly confirmed
CONTAMINATED	failed cotangent propagated upstream
PARTIALLY_CONTAMINATED	healthy and failed cotangents meet upstream
UNTESTED	no stable numerical verdict

Propagation follows reverse-mode direction. A failed UVS cotangent affects Optics; it does not make the forward sibling outputs failed. Local directional checks remain the final attribution mechanism.

Forward identical. One backward defect.

WingSpectrum is a reduced-order spectral glazing inverse-design demonstration. Ten smooth controls generate reflectance and transmittance over 300-1100 nm with fixed absorptance and $R+T+A=1$. Optics fans out to UVS, VS, Human, and Solar branches.

Frozen setting	Value
Initial design	10 x -3.0
Optimizer / rate	Adam / 0.16
Iterations / seed	180 / 20260822
Contract tolerance	rtol 1e-4, atol 1e-7
Only difference	UVS wavelength cotangent order

Real Tesseract execution

GitHub Actions builds six Tesseract Core 1.11.0 images: Optics, faulty UVS, fixed UVS, VS, Human, and Thermal. The orchestrator calls each image's apply and vector_jacobian_product endpoints, aggregates branch cotangents, applies the Optics VJP, and performs identical Adam updates.

The faulty UVS image computes the same forward visibility but reverses its 81-sample wavelength-axis cotangent. This is a silent, shape-correct backward defect.

Recorded evidence: settings JSON; every iteration's loss and branch metrics; initial/final spectra; directional checks; DAG states; endpoint evidence; exit codes; image SHAs; workflow URL.

Diagnosis changes the design outcome.



Fixed versus faulty: UVS +5.1%, VS +7.5%, and the common forward-evaluated loss improves.

Contract evidence	Faulty	Fixed
UVS status	FAIL	PASS
Optics	PARTIAL	PASS
Healthy siblings	PASS	PASS
UVS relative error	0.7103	4.25e-12
Exit code	1	0
Human reflectance	0.1388	0.1494

Reproduce

```
tesstrace --json reports/faulty.json # exit 1
tesstrace --fixed --json reports/fixed.json # exit 0
```

Verified workflow: github.com/lsh2546/tesstrace-tesseract-hackathon-2026/actions/runs/32576243257
Artifact: wingspectrum-container-comparison-32576243257

Scientific limits

No claim of measured bird-collision reduction, product-ready coating, fabrication feasibility, or improved solar/thermal performance. UVS and VS outputs are model-based spectral proxies. The optics model is a differentiable surrogate. Directional finite differences may be inconclusive near non-smooth or unstable points; TessTrace reports UNTESTED rather than forcing a verdict.